

TEST CASES DOCUMENT

Personal Finance & Expense Manager (PFEM)

Version 1.0

Date: 02-10-2026

Team 9 — PES University

Status: Draft / Ready for Execution

Standalone Test Case Deliverable derived from the PFEM Software Requirements Specification and Test Plan.

1. Purpose

This document is the standalone Test Cases deliverable for the Personal Finance & Expense Manager (PFEM). It contains the manual Unit (UT), Integration (IT), and System (ST) test cases defined for Release 1.0. The cases are derived from the approved PFEM SRS and the project Test Plan.

2. Test Case ID Convention

UT_xx = Unit Test; IT_xx = Integration Test; ST_xx = System Test.
Actual Result and Test Result are intentionally left blank so the document can be used during execution. A test is marked Pass only when the observed result matches the Expected Result.

3. Coverage Summary

Test Level	Number of Cases	Purpose
Unit (UT)	64	Module-level validation
Integration (IT)	34	Interaction between services/components
System (ST)	50	End-to-end functional, safety, security and performance validation
Total	148	Complete PFEM test-case set

4. Test Case Fields

Each case contains: Test Case ID, Module, Test Case Description, Preconditions, Test Steps, Test Data, Expected Result, Actual Result, and Test Result.

5.1 User Registration & Authentication

Test Case ID	UT_01
Module	User Registration & Authentication
Test Case Description	Verify registration succeeds with valid required fields (REQ-1)
Preconditions	User is on Registration screen
Test Steps	1. Enter full name, valid email and strong password. 2. Leave mobile optional. 3. Submit.
Test Data	Name: Girish G N; Email: user1@test.com; Password: Test@1234; Mobile: blank
Expected Result	Registration succeeds and account enters PENDING_VERIFICATION state; verification code is sent.
Actual Result	
Test Result	

Test Case ID	UT_02
Module	User Registration & Authentication
Test Case Description	Verify optional mobile number is accepted during registration (REQ-1)
Preconditions	User is on Registration screen
Test Steps	1. Enter valid required fields. 2. Enter a valid mobile number. 3. Submit.
Test Data	Mobile: 9876543210
Expected Result	Registration succeeds without rejecting the optional mobile number.
Actual Result	
Test Result	

Test Case ID	UT_03
Module	User Registration & Authentication
Test Case Description	Verify invalid email format is rejected (REQ-2)
Preconditions	Registration form is open
Test Steps	1. Enter an invalid email. 2. Submit.
Test Data	Email: girish@@test
Expected Result	Message "Enter a valid email address." is shown and registration is not submitted.
Actual Result	
Test Result	

Test Case ID	UT_04
Module	User Registration & Authentication
Test Case Description	Verify password complexity rules and unmet-rule feedback (REQ-3)
Preconditions	Registration form is open
Test Steps	1. Enter a password missing required character classes. 2. Submit.
Test Data	Password: password
Expected Result	Password is rejected and the unmet password rules are listed.
Actual Result	
Test Result	

Test Case ID	UT_05
Module	User Registration & Authentication
Test Case Description	Verify six-digit OTP format and ten-minute validity (REQ-4)
Preconditions	A registration is pending verification and OTP has been sent
Test Steps	1. Enter the received OTP within 10 minutes. 2. Submit.
Test Data	OTP: six numeric digits received by email
Expected Result	Correct six-digit OTP within 10 minutes verifies the email successfully.
Actual Result	
Test Result	

Test Case ID	UT_06
Module	User Registration & Authentication
Test Case Description	Verify OTP attempt and resend limits (REQ-4)
Preconditions	A registration is pending verification
Test Steps	1. Enter wrong OTP three times. 2. Attempt a fourth verification. 3. Request resend three times within one hour. 4. Attempt another resend.
Test Data	Wrong OTPs; resend requests within 1 hour
Expected Result	Verification does not allow more than three attempts, and resend requests are limited to three per hour.
Actual Result	
Test Result	

Test Case ID	IT_01
Module	User Registration & Authentication
Test Case Description	Verify five failed sign-ins lock the account for 15 minutes and notify owner (REQ-8)
Preconditions	An active verified account exists
Test Steps	1. Submit five consecutive wrong passwords. 2. Try the correct password during lock. 3. Check account email.
Test Data	Correct password after 5 failures
Expected Result	Account is locked for 15 minutes; even the correct password is refused during lock; owner receives email notification.
Actual Result	
Test Result	

Test Case ID	ST_01
Module	User Registration & Authentication
Test Case Description	Verify successful login issues a 24-hour JWT (REQ-7)
Preconditions	Verified active account exists
Test Steps	1. Enter valid email/password. 2. Sign in. 3. Inspect authenticated session.
Test Data	Email: user1@test.com; Password: Test@1234
Expected Result	Login succeeds and a JWT is issued with 24-hour validity.
Actual Result	
Test Result	

Test Case ID	UT_76
Module	User Registration & Authentication
Test Case Description	Verify duplicate active-account email is rejected without revealing other account details (REQ-5)
Preconditions	An active account already uses the email address
Test Steps	1. Attempt registration with the same email. 2. Inspect response and messages.
Test Data	Email: user1@test.com
Expected Result	Registration is rejected; response does not reveal details of the existing account.
Actual Result	
Test Result	

Test Case ID	UT_61
Module	User Registration & Authentication
Test Case Description	Verify password is stored only as salted bcrypt hash and never exposed as plaintext (REQ-6)
Preconditions	Test environment permits inspection of persisted user record and logs
Test Steps	1. Register a user. 2. Inspect password storage and application logs. 3. Check UI/API responses for password value.
Test Data	Password: Test@1234
Expected Result	Password is stored as a salted bcrypt hash with work factor ≥ 12 ; plaintext password is absent from storage, logs and displayed responses.
Actual Result	
Test Result	

Test Case ID	UT_62
Module	User Registration & Authentication
Test Case Description	Verify password reset link is single-use and expires after 15 minutes (REQ-9)
Preconditions	Verified user can request password reset
Test Steps	1. Request reset. 2. Use link within 15 minutes and set a new password. 3. Reuse the same link. 4. Try an expired link.
Test Data	Reset link; new password; expired link
Expected Result	Valid link works once; reused or expired link is rejected.
Actual Result	
Test Result	

Test Case ID	IT_17
Module	User Registration & Authentication
Test Case Description	Verify changing password invalidates all existing sessions (REQ-9)
Preconditions	User is logged in on two sessions/devices
Test Steps	1. Change password in session A using current password. 2. Use old session B. 3. Sign in again with new password.
Test Data	Old and new passwords
Expected Result	All existing sessions are invalidated after password change; new password can create a fresh session.
Actual Result	
Test Result	

Test Case ID	UT_63
Module	User Registration & Authentication
Test Case Description	Verify sign-out invalidates server-side session token (REQ-10)
Preconditions	User is signed in
Test Steps	1. Sign out. 2. Reuse the previous token for an authenticated request.
Test Data	Previously valid session token
Expected Result	Sign-out succeeds and the old token cannot be reused.
Actual Result	
Test Result	

Test Case ID	UT_64
Module	User Registration & Authentication
Test Case Description	Verify optional app lock uses PIN/biometric and re-locks after two minutes in background (REQ-11)
Preconditions	User is signed in and app lock is enabled
Test Steps	1. Configure 4-digit PIN or biometric. 2. Send app to background for at least two minutes. 3. Return to app.
Test Data	PIN: 4826
Expected Result	App requires configured PIN/biometric after returning from background.
Actual Result	
Test Result	

Test Case ID	UT_65
Module	User Registration & Authentication
Test Case Description	Verify Terms and Privacy consent is mandatory at registration (REQ-12)
Preconditions	Registration form is open
Test Steps	1. Leave Terms/Privacy consent unchecked. 2. Submit. 3. Repeat with consent checked.
Test Data	Consent: No, then Yes
Expected Result	No account is created without explicit consent; with consent, registration proceeds and consent is recorded.
Actual Result	
Test Result	

Test Case ID	UT_07
Module	User Registration & Authentication
Test Case Description	Verify authentication unit logic accepts valid credentials for an active verified user (REQ-7)
Preconditions	Verified active account exists
Test Steps	1. Submit valid email and password. 2. Validate authentication result.
Test Data	Email: user1@test.com; Password: Test@1234
Expected Result	Authentication succeeds for valid credentials.
Actual Result	
Test Result	

5.2 Account & Wallet Management

Test Case ID	UT_08
Module	Account & Wallet Management
Test Case Description	Verify account creation with mandatory name and type (REQ-13)
Preconditions	User is signed in on Accounts screen
Test Steps	1. Add account. 2. Enter name, type, opening balance and currency. 3. Save.
Test Data	Name: HDFC Savings; Type: Bank; Opening: ₹5000; Currency: INR
Expected Result	Account is created with the supplied values.
Actual Result	
Test Result	

Test Case ID	UT_09
Module	Account & Wallet Management
Test Case Description	Verify all supported account types and currency are accepted (REQ-13)
Preconditions	User is on Add Account
Test Steps	1. Create Cash, Bank, Credit Card and Wallet accounts. 2. Save each.
Test Data	Types: Cash, Bank, Credit Card, Wallet
Expected Result	Each supported account type can be created; name and type remain mandatory.
Actual Result	
Test Result	

Test Case ID	UT_10
Module	Account & Wallet Management
Test Case Description	Verify duplicate account name is rejected (REQ-14)
Preconditions	User already has an account named HDFC Savings
Test Steps	1. Add another account with the same name. 2. Save.
Test Data	Name: HDFC Savings
Expected Result	Duplicate name is rejected with an explanatory message.
Actual Result	
Test Result	

Test Case ID	UT_66
Module	Account & Wallet Management
Test Case Description	Verify opening-balance validation for zero, credit-card outstanding and over-precise values (REQ-15)
Preconditions	User is on Add Account
Test Steps	1. Create asset account with zero balance. 2. Create credit card with outstanding amount. 3. Enter nonnumeric and >2 decimal values.
Test Data	Cash balance: 0; Credit Card outstanding: ₹12000; Invalid: abc, 10.123
Expected Result	Zero is accepted for Cash/Bank/Wallet; credit-card outstanding is accepted; nonnumeric and >2 decimal values are rejected.
Actual Result	
Test Result	

Test Case ID	UT_67
Module	Account & Wallet Management
Test Case Description	Verify current balance cannot be edited directly (REQ-16)
Preconditions	An account with transactions exists
Test Steps	1. Open Edit Account. 2. Try to change current balance. 3. Save other editable fields.
Test Data	Attempt balance: ■99999; Name change
Expected Result	Current balance is not directly editable; only permitted account fields are changed.
Actual Result	
Test Result	

Test Case ID	UT_68
Module	Account & Wallet Management
Test Case Description	Verify account archive hides it from transaction entry but retains it in reports (REQ-17)
Preconditions	An account exists and may have historical transactions
Test Steps	1. Archive account. 2. Open transaction-entry account selector. 3. Open historical report.
Test Data	Account: Cash Wallet
Expected Result	Archived account is hidden from transaction entry and remains visible in historical reports.
Actual Result	
Test Result	

Test Case ID	UT_12
Module	Account & Wallet Management
Test Case Description	Verify deletion is allowed only for an account with no transactions (REQ-18)
Preconditions	User has one empty account and one account with transactions
Test Steps	1. Delete empty account with confirmation. 2. Try deleting account with transactions.
Test Data	Empty account; account with 2 transactions
Expected Result	Empty account can be deleted after explicit confirmation; account with transactions cannot be deleted and archive is offered.
Actual Result	
Test Result	

Test Case ID	IT_02
Module	Account & Wallet Management
Test Case Description	Verify deletion with existing transactions is blocked and archive alternative is offered (REQ-18)
Preconditions	Account has transactions
Test Steps	1. Select Delete. 2. Confirm.
Test Data	Account with 2 transactions
Expected Result	Deletion is refused; the user is offered archiving and transactions remain intact.
Actual Result	
Test Result	

Test Case ID	IT_03
Module	Account & Wallet Management
Test Case Description	Verify balance recomputes within two seconds after transaction change (REQ-19)
Preconditions	Account has known opening balance and transactions
Test Steps	1. Add/edit/delete a transaction. 2. Measure balance refresh time.
Test Data	Opening ■5000; expense ■500
Expected Result	Balance equals opening + income - expense +/- transfers and refreshes within 2 seconds.
Actual Result	
Test Result	

Test Case ID	UT_13
Module	Account & Wallet Management
Test Case Description	Verify consolidated net worth calculation (REQ-20)
Preconditions	User has asset and credit-card accounts
Test Steps	1. Create balances/outstanding. 2. Open net-worth view.
Test Data	Bank ■50000; Cash ■10000; Credit Card outstanding ■5000
Expected Result	Net worth is assets minus credit-card outstanding, shown in base currency.
Actual Result	
Test Result	

Test Case ID	ST_45
Module	Account & Wallet Management
Test Case Description	Verify credit-card details and 80% utilization warning (REQ-21)
Preconditions	Credit Card account exists with credit limit
Test Steps	1. Open credit-card account. 2. Set outstanding above 80% of limit.
Test Data	Limit ■100000; outstanding ■85000
Expected Result	Outstanding, credit limit, available credit and statement due date are displayed; warning appears above 80% of limit.
Actual Result	
Test Result	

Test Case ID	ST_02
Module	Account & Wallet Management
Test Case Description	Verify account creation, uniqueness, deletion rule, balance computation and net worth end-to-end (REQ-13, REQ-14, REQ-18, REQ-19, REQ-20)
Preconditions	User is signed in with account test data
Test Steps	1. Create valid accounts. 2. Attempt duplicate name. 3. Add transactions. 4. Verify balance/net worth. 5. Try deleting account with transactions.
Test Data	Bank ■5000; Cash ■1000; duplicate Bank name; expense ■500
Expected Result	Account creation works; duplicate names are rejected; balances/net worth are correct; account with transactions cannot be deleted.
Actual Result	
Test Result	

5.3 Transaction Management

Test Case ID	UT_14
Module	Transaction Management
Test Case Description	Verify expense accepts all mandatory and optional fields (REQ-22)
Preconditions	User is on Add Expense
Test Steps	1. Enter amount/date/account/category/payment mode. 2. Add note, tags and receipt. 3. Save.
Test Data	■500; current date; Bank; Food; UPI; note; 2 tags; receipt.pdf
Expected Result	Expense is recorded with supplied fields.
Actual Result	
Test Result	

Test Case ID	UT_15
Module	Transaction Management
Test Case Description	Verify income increases selected account balance (REQ-23)
Preconditions	User has an active account
Test Steps	1. Select Income. 2. Enter valid fields. 3. Save.
Test Data	Income ■10000 to Bank account
Expected Result	Income is recorded and selected account balance increases by the income amount.
Actual Result	
Test Result	

Test Case ID	IT_05
Module	Transaction Management
Test Case Description	Verify own-account transfer creates linked debit/credit and does not change net worth (REQ-24)
Preconditions	Two accounts belong to same user
Test Steps	1. Create transfer from Account A to B. 2. Check linked records, balances, reports and budgets.
Test Data	Transfer ■2000 from Bank to Wallet
Expected Result	Single linked transfer debits source and credits destination; net worth unchanged; transfer excluded from income, expense and budget totals.
Actual Result	
Test Result	

Test Case ID	UT_16
Module	Transaction Management
Test Case Description	Verify invalid transaction amounts are rejected with exact message (REQ-25)
Preconditions	User is on Add Transaction
Test Steps	1. Enter zero. 2. Repeat with negative, nonnumeric and >2 decimal amounts. 3. Save.
Test Data	0; -100; abc; 10.123
Expected Result	Each invalid value is rejected with "Enter an amount greater than 0 with up to 2 decimal places."
Actual Result	
Test Result	

Test Case ID	UT_69
Module	Transaction Management
Test Case Description	Verify valid amount with at most two decimals is accepted (REQ-25)
Preconditions	User is on Add Transaction
Test Steps	1. Enter positive amount with two decimals. 2. Save.
Test Data	Amount: ■1250.50
Expected Result	Transaction is accepted and stored with two decimal precision.
Actual Result	
Test Result	

Test Case ID	UT_17
Module	Transaction Management
Test Case Description	Verify transaction date before account opening is rejected (REQ-26)
Preconditions	Account opening date is known
Test Steps	1. Create transaction dated before opening. 2. Save.
Test Data	Opening: 10-Aug-2026; Transaction: 09-Aug-2026
Expected Result	Transaction is rejected with an explanatory date message.
Actual Result	
Test Result	

Test Case ID	UT_70
Module	Transaction Management
Test Case Description	Verify transaction date more than one day in future is rejected (REQ-26)
Preconditions	User is adding a transaction
Test Steps	1. Enter date more than one day after current date. 2. Save.
Test Data	Transaction date: current date + 2 days
Expected Result	Transaction is rejected with an explanatory future-date message.
Actual Result	
Test Result	

Test Case ID	UT_71
Module	Transaction Management
Test Case Description	Verify note and tag limits with live note counter (REQ-27)
Preconditions	User is on Add Transaction
Test Steps	1. Enter a 200-character note. 2. Try 201 characters. 3. Add 5 tags, then a sixth. 4. Enter a tag over 20 characters.
Test Data	Note: 200/201 chars; tags: 5 valid + sixth; one 21-char tag
Expected Result	200-character note is accepted with live counter; 201st character is blocked/rejected; max 5 tags and 20 characters per tag enforced.
Actual Result	
Test Result	

Test Case ID	UT_19
Module	Transaction Management
Test Case Description	Verify editing a transaction recomputes dependent figures immediately (REQ-28)
Preconditions	Transaction affects account, budget and report
Test Steps	1. Edit amount/category/account/date. 2. Save. 3. Observe balance, budget and report.
Test Data	Expense ■500 changed to ■800
Expected Result	Affected account balance, budget utilisation and report figures update immediately after save.
Actual Result	
Test Result	

Test Case ID	UT_20
Module	Transaction Management
Test Case Description	Verify soft delete and restore window for a transaction (REQ-29)
Preconditions	Transaction exists
Test Steps	1. Delete transaction. 2. Check balance/recycle bin. 3. Restore within 30 days. 4. Check balance again.
Test Data	Expense ■500
Expected Result	Transaction moves to recycle bin; balance is restored immediately; transaction is restorable within 30 days.
Actual Result	
Test Result	

Test Case ID	ST_13
Module	Transaction Management
Test Case Description	Verify transaction is permanently purged after 30 days and cannot be restored (REQ-29)
Preconditions	A deleted transaction is older than 30 days in test environment
Test Steps	1. Run/inspect purge process. 2. Open recycle bin. 3. Attempt restore.
Test Data	Deleted transaction dated >30 days ago
Expected Result	Transaction is permanently purged after the 30-day restore period and is no longer restorable.
Actual Result	
Test Result	

Test Case ID	ST_14
Module	Transaction Management
Test Case Description	Verify transaction list ordering, pagination and displayed fields (REQ-30)
Preconditions	More than 25 transactions exist
Test Steps	1. Open transaction list. 2. Check order and page size. 3. Inspect a row.
Test Data	30 transactions with varied dates
Expected Result	Transactions are reverse chronological; 25 appear per page or progressive loading is used; date, category, account, note and amount are shown.
Actual Result	
Test Result	

Test Case ID	ST_46
Module	Transaction Management
Test Case Description	Verify all transaction filters and note-text search return filtered total (REQ-31)
Preconditions	Transactions cover varied dates, accounts, categories, types, tags and amounts
Test Steps	1. Apply date, account, category, type, tag and amount filters. 2. Search note text. 3. Observe total.
Test Data	Multiple transactions with known attributes
Expected Result	Each filter/search narrows results correctly and filtered total is displayed.
Actual Result	
Test Result	

Test Case ID	UT_72
Module	Transaction Management
Test Case Description	Verify expense split across up to five categories sums exactly (REQ-32)
Preconditions	User is adding an expense
Test Steps	1. Split one expense across 5 categories. 2. Save. 3. Try a split whose parts do not sum to total.
Test Data	Total ■1000; split ■200+■200+■200+■200+■200; invalid total ■999
Expected Result	Exact split is accepted; non-matching split is rejected.
Actual Result	
Test Result	

Test Case ID	ST_48
Module	Transaction Management
Test Case Description	Verify excluded transaction remains in balance/reports but not budget utilisation (REQ-33)
Preconditions	Budget and transaction exist
Test Steps	1. Mark transaction excluded from budget. 2. Refresh budget, balance and report.
Test Data	Expense ■1000 excluded
Expected Result	Budget utilisation excludes the transaction, while account balance and reports still include it.
Actual Result	
Test Result	

Test Case ID	ST_47
Module	Transaction Management
Test Case Description	Verify offline transaction synchronises within 60 seconds after connectivity returns (REQ-34)
Preconditions	Mobile client is offline
Test Steps	1. Create transaction offline. 2. Confirm pending status. 3. Restore connectivity. 4. Observe sync.
Test Data	Offline expense ■250
Expected Result	Transaction is marked pending offline and synchronises to server within 60 seconds after connectivity is restored.
Actual Result	
Test Result	

Test Case ID	IT_04
Module	Transaction Management
Test Case Description	Verify transaction edit updates account, budget and report figures through integrated services (REQ-28)
Preconditions	Transaction affects account, budget and report
Test Steps	1. Edit transaction amount. 2. Save. 3. Check all dependent views.
Test Data	■500 changed to ■800
Expected Result	All dependent figures are recomputed consistently after edit.
Actual Result	
Test Result	

Test Case ID	ST_03
Module	Transaction Management
Test Case Description	Verify end-to-end expense creation, validation, transfer, edit and soft-delete behaviour (REQ-22, REQ-25, REQ-26, REQ-24, REQ-28, REQ-29)
Preconditions	Accounts and transaction test data exist
Test Steps	1. Record expense. 2. Try invalid amount/date. 3. Create own-account transfer. 4. Edit expense. 5. Soft-delete and restore.
Test Data	Expense ■500; invalid amount 0; transfer ■2000
Expected Result	Valid transaction is recorded; invalid values are rejected; transfer leaves net worth unchanged; edits update dependent figures; deleted transaction is recoverable within 30 days.
Actual Result	
Test Result	

5.4 Category & Tag Management

Test Case ID	UT_21
Module	Category & Tag Management
Test Case Description	Verify custom category creation with icon and colour (REQ-36)
Preconditions	User is on Categories
Test Steps	1. Add custom category. 2. Enter name/type/icon/colour. 3. Save.
Test Data	Name: Dining Out; Type: Expense; Icon: fork; Colour: chosen UI colour
Expected Result	Custom category is created with supplied attributes.
Actual Result	
Test Result	

Test Case ID	UT_22
Module	Category & Tag Management
Test Case Description	Verify custom category name length limit (REQ-36)
Preconditions	User is creating a custom category
Test Steps	1. Enter 30-character name. 2. Save. 3. Try 31 characters.
Test Data	30-char valid name; 31-char invalid name
Expected Result	Name up to 30 characters is accepted; longer name is rejected.
Actual Result	
Test Result	

Test Case ID	UT_73
Module	Category & Tag Management
Test Case Description	Verify default expense and income categories are created for new users (REQ-35)
Preconditions	A new user has completed registration
Test Steps	1. Sign in. 2. Open category list.
Test Data	New user
Expected Result	Default categories include at least Food, Groceries, Transport, Rent, Utilities, Health, Education, Entertainment, Shopping, Salary, Interest and Other.
Actual Result	
Test Result	

Test Case ID	UT_74
Module	Category & Tag Management
Test Case Description	Verify blank, whitespace and duplicate category names are rejected (REQ-37)
Preconditions	User is on category creation
Test Steps	1. Try blank name. 2. Try whitespace-only. 3. Create duplicate within same type.
Test Data	Blank; spaces; duplicate Food expense category
Expected Result	Invalid names are rejected with explanatory feedback; duplicate within same type is rejected.
Actual Result	
Test Result	

Test Case ID	IT_21
Module	Category & Tag Management
Test Case Description	Verify one-level subcategory rule (REQ-38)
Preconditions	Parent category exists
Test Steps	1. Create a subcategory under a category. 2. Try creating a subcategory under that subcategory.
Test Data	Food > Dining; Dining > Fast Food
Expected Result	First-level subcategory is allowed; second-level nesting is rejected.
Actual Result	
Test Result	

Test Case ID	IT_22
Module	Category & Tag Management
Test Case Description	Verify system-defined categories cannot be renamed/deleted (REQ-39)
Preconditions	Default system category exists
Test Steps	1. Open edit/delete for Food. 2. Attempt rename/delete.
Test Data	System category: Food
Expected Result	System-defined category cannot be renamed or deleted.
Actual Result	
Test Result	

Test Case ID	UT_25
Module	Category & Tag Management
Test Case Description	Verify custom category deletion reassigns transactions to Uncategorized (REQ-40)
Preconditions	Custom category has existing transactions
Test Steps	1. Delete category. 2. Review confirmation. 3. Confirm.
Test Data	Category: Dining Out; 3 linked transactions
Expected Result	Confirmation states 3 affected transactions; after confirmation all are reassigned to Uncategorized.
Actual Result	
Test Result	

Test Case ID	ST_49
Module	Category & Tag Management
Test Case Description	Verify tag suggestions appear while typing and tag filter works (REQ-41)
Preconditions	Transactions with previously used tags exist
Test Steps	1. Start typing an existing tag. 2. Select suggestion. 3. Filter transaction list by tag.
Test Data	Tag: vacation
Expected Result	Previously used tags are suggested; tag filter returns matching transactions.
Actual Result	
Test Result	

Test Case ID	ST_50
Module	Category & Tag Management
Test Case Description	Verify category-wise and tag-wise totals are correct and refresh after changes (REQ-42)
Preconditions	Transactions have categories and tags
Test Steps	1. Open totals for a period. 2. Add/edit/delete a tagged transaction. 3. Observe totals.
Test Data	Known category/tag amounts
Expected Result	Category-wise and tag-wise totals are correct and update within five seconds after add/edit/delete.
Actual Result	
Test Result	

Test Case ID	IT_06
Module	Category & Tag Management
Test Case Description	Verify category-wise and tag-wise totals through report aggregation (REQ-42)
Preconditions	Transactions with categories and tags exist
Test Steps	1. Select a period. 2. Compare category and tag totals with transaction data.
Test Data	Known tagged/category amounts
Expected Result	Category-wise and tag-wise totals match transaction data.
Actual Result	
Test Result	

Test Case ID	ST_04
Module	Category & Tag Management
Test Case Description	Verify end-to-end custom category creation, deletion and aggregation (REQ-36, REQ-40, REQ-42)
Preconditions	User has transactions and tags
Test Steps	1. Create custom category. 2. Attach to transactions. 3. Delete category. 4. Inspect Uncategorized and totals.
Test Data	Dining Out category; 3 linked transactions; tag vacation
Expected Result	Custom category works; deletion shows affected count and reassigns transactions to Uncategorized; category/tag totals remain correct.
Actual Result	
Test Result	

5.5 Budget Management & Alerts

Test Case ID	UT_27
Module	Budget Management & Alerts
Test Case Description	Verify positive budget creation for category and overall spending (REQ-43)
Preconditions	User is on Budgets
Test Steps	1. Create category budget. 2. Create overall-spending budget. 3. Save.
Test Data	Food monthly ■6000; overall weekly ■5000
Expected Result	Budgets with amount >0 are created for supported categories/overall and weekly/monthly/custom periods.
Actual Result	
Test Result	

Test Case ID	UT_28
Module	Budget Management & Alerts
Test Case Description	Verify zero and negative budgets are rejected (REQ-43)
Preconditions	User is on Add Budget
Test Steps	1. Enter 0. 2. Enter negative value. 3. Save.
Test Data	Amount: 0 and -500
Expected Result	Budget is rejected because amount must be greater than zero.
Actual Result	
Test Result	

Test Case ID	UT_29
Module	Budget Management & Alerts
Test Case Description	Verify overlapping budget for same category and period is rejected (REQ-44)
Preconditions	A Food monthly budget already exists
Test Steps	1. Create another overlapping Food monthly budget. 2. Save.
Test Data	Food; same month; ■7000
Expected Result	Message "A budget already exists for this category in this period." is shown.
Actual Result	
Test Result	

Test Case ID	ST_15
Module	Budget Management & Alerts
Test Case Description	Verify active budget displays all required utilisation fields (REQ-45)
Preconditions	Active budget exists
Test Steps	1. Open budget. 2. Inspect values.
Test Data	Budget ■6000; spent ■2400; period with known days left
Expected Result	Amount budgeted, amount spent, remaining, percentage used and days left are all displayed correctly.
Actual Result	
Test Result	

Test Case ID	IT_07
Module	Budget Management & Alerts
Test Case Description	Verify 80% warning is raised once (REQ-46)
Preconditions	Budget exists below 80%
Test Steps	1. Add spending reaching exactly 80%. 2. Add more spending while still above 80%.
Test Data	Budget ■6000; spend to ■4800 then +■50
Expected Result	80% warning is raised once for the budget period and not duplicated by later spending.
Actual Result	
Test Result	

Test Case ID	IT_08
Module	Budget Management & Alerts
Test Case Description	Verify 100% over-budget alert is raised once (REQ-46)
Preconditions	Budget exists below 100%
Test Steps	1. Add spending to/exceeding 100%. 2. Add more spending.
Test Data	Budget ■6000; spend ■6100 then +■50
Expected Result	Over-budget alert is raised once at >=100%; later spending does not create duplicate alert for the same period.
Actual Result	
Test Result	

Test Case ID	ST_16
Module	Budget Management & Alerts
Test Case Description	Verify budget edit recalculates utilisation and deletion preserves transactions (REQ-47)
Preconditions	Budget has linked transactions
Test Steps	1. Edit budget amount. 2. Observe utilisation. 3. Delete budget. 4. Inspect transactions.
Test Data	Budget ■6000 changed to ■7000; linked expenses
Expected Result	Utilisation recalculates immediately on edit; deleting budget does not delete or alter transactions.
Actual Result	
Test Result	

Test Case ID	UT_32
Module	Budget Management & Alerts
Test Case Description	Verify rollover carries unused amount and displays it separately (REQ-48)
Preconditions	Budget rollover is enabled
Test Steps	1. Finish period with unused amount. 2. Open next period.
Test Data	Base ■6000; spent ■4000; unused ■2000
Expected Result	Next period shows base ■6000 plus separate rollover ■2000.
Actual Result	
Test Result	

Test Case ID	UT_75
Module	Budget Management & Alerts
Test Case Description	Verify repeat setting creates equivalent budget for next period (REQ-49)
Preconditions	Budget repeat option is available
Test Steps	1. Enable Repeat. 2. End current period. 3. Open next period.
Test Data	Food monthly ■6000; Repeat enabled
Expected Result	Equivalent Food budget is automatically created for the next period.
Actual Result	
Test Result	

Test Case ID	ST_17
Module	Budget Management & Alerts
Test Case Description	Verify excluded transactions and own-account transfers are excluded from budget utilisation (REQ-50)
Preconditions	Budget exists; excluded transaction and transfer exist
Test Steps	1. Add excluded expense. 2. Transfer between own accounts. 3. Observe budget utilisation.
Test Data	Excluded expense ■500; transfer ■2000
Expected Result	Neither excluded transaction nor own-account transfer contributes to budget utilisation.
Actual Result	
Test Result	

Test Case ID	ST_56
Module	Budget Management & Alerts
Test Case Description	Verify utilisation refreshes within five seconds after add, edit, delete and import (REQ-51)
Preconditions	Active budget exists
Test Steps	1. Add transaction. 2. Edit it. 3. Delete it. 4. Import a valid transaction. 5. Measure each refresh.
Test Data	Known budget and transactions
Expected Result	Budget utilisation refreshes within five seconds after each add/edit/delete/import.
Actual Result	
Test Result	

Test Case ID	IT_36
Module	Budget Management & Alerts
Test Case Description	Verify 100% budget alert is raised when spending reaches or exceeds budget (REQ-46)
Preconditions	Budget exists below 100%
Test Steps	1. Add spending to 100% or above. 2. Observe notification.
Test Data	Budget ■6000; spend ■6100
Expected Result	Over-budget alert is raised once when spending reaches or exceeds 100%.
Actual Result	
Test Result	

Test Case ID	ST_05
Module	Budget Management & Alerts
Test Case Description	Verify end-to-end budget creation, overlap validation, alerts and rollover (REQ-43, REQ-44, REQ-46, REQ-48)
Preconditions	Budgeted category has transactions
Test Steps	1. Create budget. 2. Attempt overlap. 3. Spend to 80% and 100%. 4. End period with unused amount and rollover enabled.
Test Data	Food budget ■6000; spend ■4800 then ■6100; unused ■2000
Expected Result	Valid budget is created; overlap is rejected; 80% and 100% alerts are raised once; rollover is carried separately to next period.
Actual Result	
Test Result	

5.6 Recurring Transactions & Bill Reminders

Test Case ID	UT_33
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify recurring rule creation for supported frequency (REQ-52)
Preconditions	User is on Recurring screen
Test Steps	1. Add rule with type, amount, account, category, frequency and start date. 2. Save.
Test Data	Expense ■15000; Monthly; start 01-Oct-2026
Expected Result	Rule is saved and next due date is shown.
Actual Result	
Test Result	

Test Case ID	UT_34
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify recurring rule supports end date, occurrence count or neither (REQ-52)
Preconditions	User is creating recurring rules
Test Steps	1. Create rules using end date, occurrence count and no end condition. 2. Save.
Test Data	End date; count 12; neither
Expected Result	Each permitted recurrence configuration is accepted.
Actual Result	
Test Result	

Test Case ID	UT_18
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify end date before start date is rejected with exact message (REQ-53)
Preconditions	User is creating recurring rule
Test Steps	1. Enter start date. 2. Enter earlier end date. 3. Save.
Test Data	Start 01-Oct-2026; End 01-Sep-2026
Expected Result	Message "End date must be after the start date." is displayed.
Actual Result	
Test Result	

Test Case ID	IT_09
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify scheduler generates due transaction at 00:05 IST and marks it system-generated (REQ-54)
Preconditions	Active recurring rule is due
Test Steps	1. Run scheduler at due time. 2. Inspect generated transaction. 3. Edit generated transaction.
Test Data	Monthly Rent ■15000
Expected Result	Transaction is created at due date/time, marked system-generated, balance updated, and remains editable.
Actual Result	
Test Result	

Test Case ID	IT_23
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify nonexistent monthly day uses last day of month (REQ-55)
Preconditions	Monthly rule is configured for day 31
Test Steps	1. Process a month with fewer than 31 days. 2. Inspect due date.
Test Data	Monthly day 31; February
Expected Result	Transaction is generated on the last day of that month.
Actual Result	
Test Result	

Test Case ID	IT_24
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify pause, resume, edit and delete affect only intended future occurrences (REQ-56)
Preconditions	Recurring rule has generated transactions
Test Steps	1. Pause and resume. 2. Verify no backfill. 3. Edit rule. 4. Delete rule. 5. Inspect old generated transactions.
Test Data	Rule and future occurrences
Expected Result	Pause does not back-fill; edits affect only future occurrences; deletion retains transactions already generated.
Actual Result	
Test Result	

Test Case ID	IT_25
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify missed scheduler run recovers original dates without duplicates (REQ-60)
Preconditions	A due recurring rule was missed on a prior date
Test Steps	1. Run scheduler successfully later. 2. Inspect missed transaction date. 3. Run scheduler again same day.
Test Data	Missed due date; one recurring rule
Expected Result	Missed transaction is generated with original due date and repeated scheduler runs do not create duplicates.
Actual Result	
Test Result	

Test Case ID	UT_77
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify bill reminder creation and default notification lead time (REQ-57)
Preconditions	User is on Bill Reminders
Test Steps	1. Add reminder with name, amount, due date. 2. Leave notify-days-before blank. 3. Save.
Test Data	Bill ■3000; due 20-Oct-2026
Expected Result	Reminder is created and notify-days-before defaults to 3.
Actual Result	
Test Result	

Test Case ID	IT_28
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify bill reminder sends push and email on notification date (REQ-57)
Preconditions	Reminder exists with notification date reached
Test Steps	1. Trigger notification date. 2. Check push and email.
Test Data	Due date 20-Oct; notify 3 days before
Expected Result	Push notification and email are sent on computed notification date.
Actual Result	
Test Result	

Test Case ID	ST_18
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify marking bill paid creates linked expense and PAID state (REQ-58)
Preconditions	Unpaid reminder exists
Test Steps	1. Mark reminder as paid. 2. Inspect transactions and reminder.
Test Data	Bill █3000; account Bank
Expected Result	Linked expense transaction is created and reminder state becomes PAID.
Actual Result	
Test Result	

Test Case ID	ST_19
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify unpaid reminder becomes OVERDUE after due date (REQ-59)
Preconditions	Unpaid reminder due date has passed
Test Steps	1. Advance test date beyond due date. 2. Open dashboard/reminders.
Test Data	Due date yesterday
Expected Result	Reminder becomes OVERDUE and overdue bills are prominently displayed on dashboard.
Actual Result	
Test Result	

Test Case ID	IT_37
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify scheduler automatically generates a due recurring transaction (REQ-54)
Preconditions	Active recurring rule is due
Test Steps	1. Trigger scheduler at 00:05 IST. 2. Inspect generated transaction.
Test Data	Monthly Rent █15000
Expected Result	Due transaction is auto-created, marked system-generated, balance updated and transaction remains editable.
Actual Result	
Test Result	

Test Case ID	UT_37
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify bill reminder creation with required fields and default notification setting (REQ-57)
Preconditions	User is on Bill Reminders
Test Steps	1. Add bill reminder. 2. Leave notify-days-before at default. 3. Save.
Test Data	Name: Electricity; Amount ■3000; Due date 20-Oct-2026
Expected Result	Reminder is created and notify-days-before defaults to three.
Actual Result	
Test Result	

Test Case ID	IT_10
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify reminder notification is delivered by push and email (REQ-57)
Preconditions	Bill reminder notification date is reached
Test Steps	1. Trigger notification date. 2. Check push and email channels.
Test Data	Due 20-Oct-2026; notify 3 days before
Expected Result	Push notification and email are sent on the computed notification date.
Actual Result	
Test Result	

Test Case ID	ST_06
Module	Recurring Transactions & Bill Reminders
Test Case Description	Verify recurring transaction generation and bill reminder lifecycle end-to-end (REQ-52, REQ-54, REQ-57)
Preconditions	Recurring rule and bill reminder exist
Test Steps	1. Run scheduler. 2. Inspect recurring transaction. 3. Trigger reminder. 4. Mark bill paid. 5. Check overdue behaviour for unpaid reminder.
Test Data	Monthly Rent ■15000; bill ■3000
Expected Result	Due recurring transaction is generated correctly; reminders notify as configured; paid bill creates linked expense; unpaid bill becomes overdue.
Actual Result	
Test Result	

5.7 Savings Goals

Test Case ID	UT_39
Module	Savings Goals
Test Case Description	Verify savings goal creation with name, target, date and optional linked account (REQ-61)
Preconditions	User is on Savings Goals
Test Steps	1. Add goal. 2. Enter name, target amount and future date. 3. Optionally link account. 4. Save.
Test Data	Laptop; ■80000; 31-Mar-2027; linked Bank
Expected Result	Goal is created at 0% progress and optional linked account is stored.
Actual Result	
Test Result	

Test Case ID	UT_41
Module	Savings Goals
Test Case Description	Verify invalid target amount/date are rejected (REQ-62)
Preconditions	User is creating goal
Test Steps	1. Enter zero/negative target. 2. Enter today/past target date. 3. Save.
Test Data	Target 0, -100; target date today/past
Expected Result	Invalid target amount/date is rejected with explanatory feedback.
Actual Result	
Test Result	

Test Case ID	IT_26
Module	Savings Goals
Test Case Description	Verify contribution transfers funds, reduces source balance and is excluded from expense reports (REQ-63)
Preconditions	Goal and source account exist with sufficient balance
Test Steps	1. Contribute to goal. 2. Check source balance, goal saved amount and expense report.
Test Data	Contribution ■5000
Expected Result	Contribution is recorded as transfer; source balance decreases; goal increases; contribution is not counted as an expense.
Actual Result	
Test Result	

Test Case ID	UT_43
Module	Savings Goals
Test Case Description	Verify contribution exceeding available balance is rejected with exact message (REQ-64)
Preconditions	Source account has insufficient balance
Test Steps	1. Contribute more than available balance. 2. Save.
Test Data	Balance ■3000; contribution ■5000
Expected Result	Message "Insufficient balance in <account name>" is shown and no partial record is created.
Actual Result	
Test Result	

Test Case ID	ST_20
Module	Savings Goals
Test Case Description	Verify goal progress details are displayed accurately (REQ-65)
Preconditions	Goal has saved amount and target
Test Steps	1. Open goal details. 2. Inspect progress fields.
Test Data	Target ■80000; saved ■5000; known date
Expected Result	Saved amount, remaining amount, progress to one decimal place, days remaining and monthly saving required are displayed.
Actual Result	
Test Result	

Test Case ID	IT_11
Module	Savings Goals
Test Case Description	Verify goal becomes ACHIEVED at target and sends notification (REQ-66)
Preconditions	Goal is below target
Test Steps	1. Contribute amount that reaches/exceeds target. 2. Check state and notification. 3. Attempt another contribution.
Test Data	Target ■80000; saved ■75000; contribution ■5000
Expected Result	Goal becomes ACHIEVED, user is notified, and further contributions are rejected.
Actual Result	
Test Result	

Test Case ID	IT_27
Module	Savings Goals
Test Case Description	Verify withdrawal is limited to saved amount and recalculates progress (REQ-67)
Preconditions	Goal has saved amount
Test Steps	1. Withdraw within saved amount. 2. Check linked account and progress. 3. Attempt over-withdrawal.
Test Data	Saved ■10000; withdraw ■3000; invalid ■12000
Expected Result	Valid withdrawal returns funds to account and recalculates progress; over-withdrawal is rejected.
Actual Result	
Test Result	

Test Case ID	ST_21
Module	Savings Goals
Test Case Description	Verify goal edit and deletion with confirmation returns saved amount (REQ-68)
Preconditions	Goal with linked account and saved amount exists
Test Steps	1. Edit goal. 2. Save. 3. Delete and inspect confirmation. 4. Confirm.
Test Data	Saved ■5000; linked Bank
Expected Result	Goal edits persist; deletion requires explicit confirmation and returns saved amount to linked account.
Actual Result	
Test Result	

Test Case ID	ST_07
Module	Savings Goals
Test Case Description	Verify savings goal creation, insufficient contribution handling and achievement lifecycle (REQ-61, REQ-64, REQ-66)
Preconditions	Goal and source account exist
Test Steps	1. Create goal. 2. Attempt excessive contribution. 3. Contribute to target. 4. Inspect notification/state.
Test Data	Goal ■80000; source balance ■3000 then sufficient balance
Expected Result	Goal is created; excessive contribution is rejected without partial record; reaching target marks goal ACHIEVED and notifies user.
Actual Result	
Test Result	

5.8 Dashboard, Reports & Analytics

Test Case ID	UT_45
Module	Dashboard, Reports & Analytics
Test Case Description	Verify dashboard totals for current budget period (REQ-69)
Preconditions	User has current-period transactions and accounts
Test Steps	1. Sign in. 2. Open Dashboard.
Test Data	Known income, expenses and account balances
Expected Result	Total income, expense, net savings and consolidated net worth are displayed in base currency.
Actual Result	
Test Result	

Test Case ID	UT_46
Module	Dashboard, Reports & Analytics
Test Case Description	Verify category-wise report percentages and amounts (REQ-70)
Preconditions	Categorized expenses exist for selected period
Test Steps	1. Open report. 2. View chart/table.
Test Data	Food ■3000; Transport ■1000; Rent ■6000
Expected Result	Percentages sum to 100% and category amounts sum to period expense total.
Actual Result	
Test Result	

Test Case ID	UT_78
Module	Dashboard, Reports & Analytics
Test Case Description	Verify six-month and twelve-month income-versus-expense trends (REQ-71)
Preconditions	Transaction history covers at least 12 months
Test Steps	1. Select 6 months. 2. Select 12 months.
Test Data	12 months of monthly income/expense data
Expected Result	Line/bar trend changes to selected six- or twelve-month range and values are correct.
Actual Result	
Test Result	

Test Case ID	ST_51
Module	Dashboard, Reports & Analytics
Test Case Description	Verify top five spending categories and five largest transactions (REQ-72)
Preconditions	Varied transactions exist
Test Steps	1. Open dashboard/reports for period. 2. Inspect rankings.
Test Data	More than five categories and transactions
Expected Result	Exactly top five categories and five largest transactions are displayed in correct order.
Actual Result	
Test Result	

Test Case ID	IT_20
Module	Dashboard, Reports & Analytics
Test Case Description	Verify custom date range recomputes every figure and chart (REQ-73)
Preconditions	Transactions span multiple months
Test Steps	1. Select custom date range. 2. Apply.
Test Data	01-Jul-2026 to 15-Aug-2026
Expected Result	All report figures/charts include exactly the selected range and exclude outside data.
Actual Result	
Test Result	

Test Case ID	UT_79
Module	Dashboard, Reports & Analytics
Test Case Description	Verify defined empty state for report with no data (REQ-74)
Preconditions	User has no transactions in selected period
Test Steps	1. Open Reports. 2. Select empty period.
Test Data	No transactions
Expected Result	Defined guidance message is shown; no blank chart area or error is displayed.
Actual Result	
Test Result	

Test Case ID	IT_29
Module	Dashboard, Reports & Analytics
Test Case Description	Verify period comparison shows absolute and percentage differences (REQ-75)
Preconditions	Two comparable periods have data
Test Steps	1. Select Compare Periods. 2. Choose periods.
Test Data	This month vs last month
Expected Result	Total and per-category differences are shown in absolute and percentage terms, with increases/decreases distinguished.
Actual Result	
Test Result	

Test Case ID	ST_52
Module	Dashboard, Reports & Analytics
Test Case Description	Verify average daily spend and projected period-end total (REQ-76)
Preconditions	Partial-period spending data exists
Test Steps	1. Open dashboard/report. 2. Inspect projection.
Test Data	10 days elapsed; ■5000 spent
Expected Result	Average daily spend and projected period-end total are calculated and displayed.
Actual Result	
Test Result	

Test Case ID	UT_54
Module	Dashboard, Reports & Analytics
Test Case Description	Verify report PDF and CSV exports match on-screen figures (REQ-77)
Preconditions	A report is displayed
Test Steps	1. Export PDF. 2. Export underlying data as CSV. 3. Compare figures with screen.
Test Data	Monthly report
Expected Result	PDF and CSV exports are generated and exported figures/data match displayed figures.
Actual Result	
Test Result	

Test Case ID	ST_53
Module	Dashboard, Reports & Analytics
Test Case Description	Verify report generation completes within five seconds for 12 months/5000 transactions (REQ-78)
Preconditions	Dataset contains 12 months and 5000 transactions
Test Steps	1. Generate report. 2. Measure elapsed time.
Test Data	5000 transactions
Expected Result	Report generation completes within five seconds.
Actual Result	
Test Result	

Test Case ID	UT_80
Module	Dashboard, Reports & Analytics
Test Case Description	Verify category-wise expense breakdown renders correctly (REQ-70)
Preconditions	Categorized expenses exist
Test Steps	1. Open report. 2. Inspect chart and table.
Test Data	Food ■3000; Transport ■1000; Rent ■6000
Expected Result	Category breakdown chart and corresponding table are rendered correctly.
Actual Result	
Test Result	

Test Case ID	UT_81
Module	Dashboard, Reports & Analytics
Test Case Description	Verify PDF and CSV report export are available and figures match (REQ-77)
Preconditions	A report is displayed
Test Steps	1. Export PDF. 2. Export CSV. 3. Compare with screen.
Test Data	Monthly report
Expected Result	Both exports are produced and exported figures are identical to displayed figures.
Actual Result	
Test Result	

Test Case ID	ST_08
Module	Dashboard, Reports & Analytics
Test Case Description	Verify dashboard and category report figures match underlying transaction data (REQ-69, REQ-70)
Preconditions	Transactions and accounts exist
Test Steps	1. Open dashboard. 2. Open category report. 3. Compare totals and categories.
Test Data	Known income/expense dataset
Expected Result	Dashboard totals and category breakdown match the underlying data.
Actual Result	
Test Result	

Test Case ID	ST_09
Module	Dashboard, Reports & Analytics
Test Case Description	Verify report PDF and CSV exports contain the same figures as the displayed report (REQ-77)
Preconditions	A report is displayed
Test Steps	1. Export PDF. 2. Export CSV. 3. Compare figures with on-screen report.
Test Data	Monthly report
Expected Result	PDF and CSV are generated and exported figures/data match displayed figures.
Actual Result	
Test Result	

5.9 Data Import, Export & Attachments

Test Case ID	UT_51
Module	Data Import, Export & Attachments
Test Case Description	Verify up to three supported receipt attachments are accepted (REQ-79)
Preconditions	User is adding/editing a transaction
Test Steps	1. Attach one, two and three JPG/PNG/PDF files, each <=5MB. 2. Save.
Test Data	3 files: receipt1.jpg 2MB; receipt2.png 3MB; receipt3.pdf 5MB
Expected Result	All three files upload successfully and thumbnails are shown against transaction.
Actual Result	
Test Result	

Test Case ID	UT_52
Module	Data Import, Export & Attachments
Test Case Description	Verify unsupported attachment type is rejected (REQ-80)
Preconditions	User is adding/editing a transaction
Test Steps	1. Select unsupported file. 2. Upload.
Test Data	receipt.exe
Expected Result	Upload is rejected before upload where possible and permitted types are named.
Actual Result	
Test Result	

Test Case ID	UT_53
Module	Data Import, Export & Attachments
Test Case Description	Verify attachment over 5MB is rejected (REQ-80)
Preconditions	User is adding/editing a transaction
Test Steps	1. Select file >5MB. 2. Upload.
Test Data	receipt.pdf 6MB
Expected Result	Upload is rejected and size limit is communicated.
Actual Result	
Test Result	

Test Case ID	ST_54
Module	Data Import, Export & Attachments
Test Case Description	Verify attachment view/download/delete uses short-lived signed URL for owner (REQ-81)
Preconditions	Transaction owns an attachment
Test Steps	1. View/download as owner. 2. Delete. 3. Try accessing expired/other-user URL.
Test Data	Attachment receipt.pdf; signed URL
Expected Result	Owner can view/download/delete through authorized short-lived signed URL; unauthorized or expired URL does not expose file.
Actual Result	
Test Result	

Test Case ID	IT_30
Module	Data Import, Export & Attachments
Test Case Description	Verify transaction CSV export uses documented order, ISO dates and decimal amounts (REQ-82)
Preconditions	Transactions exist for selected range
Test Steps	1. Export CSV. 2. Inspect headers, dates and amounts.
Test Data	Date range with transactions
Expected Result	CSV follows documented column order, uses ISO 8601 dates and unformatted decimal amounts.
Actual Result	
Test Result	

Test Case ID	UT_55
Module	Data Import, Export & Attachments
Test Case Description	Verify CSV import preview and column mapping occur before writing data (REQ-83)
Preconditions	Valid CSV is available
Test Steps	1. Select Import. 2. Upload CSV. 3. Inspect preview. 4. Map columns.
Test Data	CSV with Date, Amount, Account, Category, Note columns
Expected Result	Preview appears before any data is written and user can map source columns to expected fields.
Actual Result	
Test Result	

Test Case ID	UT_56
Module	Data Import, Export & Attachments
Test Case Description	Verify invalid CSV rows report row number/reason and only valid rows import after confirmation (REQ-84)
Preconditions	CSV contains valid and invalid rows
Test Steps	1. Upload and preview. 2. Inspect validation errors. 3. Confirm import.
Test Data	Rows 2,4 valid; row 3 invalid amount
Expected Result	Each invalid row shows row number and reason; only valid rows are imported after confirmation.
Actual Result	
Test Result	

Test Case ID	IT_31
Module	Data Import, Export & Attachments
Test Case Description	Verify probable duplicates are flagged and skipped unless user chooses import (REQ-85)
Preconditions	CSV contains row matching existing transaction
Test Steps	1. Import CSV. 2. Inspect duplicate flag. 3. Confirm without override. 4. Repeat and choose import for duplicate.
Test Data	Matching date/amount/account/note
Expected Result	Matching row is flagged as probable duplicate and skipped by default; user can explicitly choose to import it.
Actual Result	
Test Result	

Test Case ID	IT_14
Module	Data Import, Export & Attachments
Test Case Description	Verify balances, budgets and reports update after import (REQ-86)
Preconditions	Valid import is ready
Test Steps	1. Record pre-import figures. 2. Import valid rows. 3. Compare figures.
Test Data	Imported expense ■500
Expected Result	Account balances, budget utilisation and report figures reflect imported transactions exactly as manual entries.
Actual Result	
Test Result	

Test Case ID	ST_55
Module	Data Import, Export & Attachments
Test Case Description	Verify full JSON data export contains all supported user data (REQ-87)
Preconditions	User has accounts, categories, transactions, budgets, goals, recurring rules and reminders
Test Steps	1. Request JSON export. 2. Inspect file sections.
Test Data	User with data in all supported areas
Expected Result	Single JSON file contains all of the user's supported data categories.
Actual Result	
Test Result	

Test Case ID	UT_82
Module	Data Import, Export & Attachments
Test Case Description	Verify three supported receipt attachments can be attached to one transaction (REQ-79)
Preconditions	User is adding/editing a transaction
Test Steps	1. Attach three supported files <=5MB. 2. Save transaction.
Test Data	JPG 2MB; PNG 3MB; PDF 5MB
Expected Result	Three attachments are accepted and thumbnails are displayed.
Actual Result	
Test Result	

Test Case ID	IT_38
Module	Data Import, Export & Attachments
Test Case Description	Verify post-import balances, budgets and reports recompute consistently (REQ-86)
Preconditions	Valid import is ready
Test Steps	1. Record pre-import figures. 2. Import rows. 3. Compare all dependent figures.
Test Data	Imported expense ■500
Expected Result	Balances, budgets and reports reflect imported transactions exactly as manual entry.
Actual Result	
Test Result	

Test Case ID	ST_10
Module	Data Import, Export & Attachments
Test Case Description	Verify attachments, CSV preview/import and post-import recomputation end-to-end (REQ-79, REQ-80, REQ-83, REQ-86)
Preconditions	User has transaction/import test data
Test Steps	1. Attach supported files. 2. Import CSV and preview/map. 3. Confirm valid rows. 4. Inspect balances/budget/reports.
Test Data	3 attachments; mixed-validity CSV; valid expense ■500
Expected Result	Supported attachments work; CSV preview occurs before write; only valid rows import after confirmation; balances, budgets and reports update accordingly.
Actual Result	
Test Result	

5.10 Profile, Settings, Security & Administration

Test Case ID	UT_59
Module	Profile, Settings, Security & Administration
Test Case Description	Verify password change succeeds only with correct current password (REQ-89)
Preconditions	User is on password settings
Test Steps	1. Enter correct current password and new valid password. 2. Save.
Test Data	Current: Test@1234; New: New@1234
Expected Result	Password changes successfully only after current password verification.
Actual Result	
Test Result	

Test Case ID	UT_60
Module	Profile, Settings, Security & Administration
Test Case Description	Verify incorrect current password is rejected with exact message (REQ-89)
Preconditions	User is changing password
Test Steps	1. Enter incorrect current password. 2. Submit.
Test Data	Current: Wrong@1234
Expected Result	Message "Your current password is incorrect." is displayed and password is not changed.
Actual Result	
Test Result	

Test Case ID	IT_15
Module	Profile, Settings, Security & Administration
Test Case Description	Verify profile changes require email re-verification for a new email (REQ-88)
Preconditions	User is on Profile
Test Steps	1. Change name, email, monthly income, currency and period start day. 2. Save. 3. Complete new email verification.
Test Data	New email: newuser@test.com; income █60000; INR; start day 25
Expected Result	Profile changes persist; new email becomes active only after re-verification.
Actual Result	
Test Result	

Test Case ID	IT_32
Module	Profile, Settings, Security & Administration
Test Case Description	Verify budget-breach and security alerts cannot be disabled (REQ-90)
Preconditions	User is on notification settings
Test Steps	1. Attempt to disable budget-breach alert. 2. Attempt to disable security alert. 3. Configure other reminder channel.
Test Data	Budget breach: Off; Security: Off
Expected Result	Protected budget/security alerts remain enabled; permitted reminder/summary channels can be configured.
Actual Result	
Test Result	

Test Case ID	IT_33
Module	Profile, Settings, Security & Administration
Test Case Description	Verify application lock, theme and default account settings persist across sessions/devices (REQ-91)
Preconditions	User is signed in on supported clients
Test Steps	1. Set lock, theme and default quick-entry account. 2. Sign out/in and open another device/client.
Test Data	App lock enabled; chosen theme; default Bank account
Expected Result	Configured settings persist across sessions and devices.
Actual Result	
Test Result	

Test Case ID	IT_34
Module	Profile, Settings, Security & Administration
Test Case Description	Verify changing budget-period start day recomputes figures without changing transactions (REQ-92)
Preconditions	Transactions exist across periods
Test Steps	1. Record transaction list and totals. 2. Change period start day. 3. Compare transactions and recalculated figures.
Test Data	Start day 1 changed to 25
Expected Result	Budgets/dashboard/reports recompute for new periods; stored transaction records remain unchanged.
Actual Result	
Test Result	

Test Case ID	IT_16
Module	Profile, Settings, Security & Administration
Test Case Description	Verify RBAC returns HTTP 403 for unauthorized role actions (REQ-93)
Preconditions	User and Administrator roles exist
Test Steps	1. Call endpoints allowed only to other role. 2. Inspect response. 3. Call permitted endpoint.
Test Data	User attempting admin action
Expected Result	Every request is role-checked; unauthorized role receives HTTP 403.
Actual Result	
Test Result	

Test Case ID	IT_35
Module	Profile, Settings, Security & Administration
Test Case Description	Verify administrator can manage account status but cannot view financial data (REQ-94)
Preconditions	Administrator and test user accounts exist
Test Steps	1. Admin views counts/status, suspends/restores user and audit logs. 2. Attempt to view user transaction amounts, balances, attachments or reports.
Test Data	Admin account; user account
Expected Result	Allowed admin functions work; financial details are inaccessible to administrator.
Actual Result	
Test Result	

Test Case ID	ST_11
Module	Profile, Settings, Security & Administration
Test Case Description	Verify account deletion requires password confirmation and data erasure/anonymisation within 30 days (REQ-95)
Preconditions	User account contains personal and financial data
Test Steps	1. Request deletion. 2. Confirm with password. 3. Verify confirmation email and scheduled erasure/anonymisation.
Test Data	Correct password
Expected Result	Deletion is confirmed; personal and financial data is erased or irreversibly anonymised within 30 days and completion is confirmed by email.
Actual Result	
Test Result	

Test Case ID	ST_26
Module	Profile, Settings, Security & Administration
Test Case Description	Verify required audit events are recorded and retained for 12 months (REQ-96)
Preconditions	Audit logging is enabled
Test Steps	1. Perform authentication, permission failure, export, bulk deletion and admin action. 2. Inspect audit logs. 3. Check retention boundary.
Test Data	Known actor/actions
Expected Result	Each required event records actor, action, timestamp and source address and is retained for 12 months.
Actual Result	
Test Result	

Test Case ID	ST_12
Module	Profile, Settings, Security & Administration
Test Case Description	Verify inactive web session expires after 30 minutes (SEC-8)
Preconditions	User has an authenticated web session
Test Steps	1. Remain inactive for 30 minutes. 2. Attempt an authenticated action.
Test Data	Inactive session >30 minutes
Expected Result	Session is expired and authenticated action requires sign-in again.
Actual Result	
Test Result	

Test Case ID	ST_28
Module	Profile, Settings, Security & Administration
Test Case Description	Verify automated encrypted database backups occur every 24 hours and retain seven days (REQ-97)
Preconditions	Backup scheduler is configured in test environment
Test Steps	1. Inspect backup schedule and encryption. 2. Verify seven-day retention. 3. Restore to selected retained point.
Test Data	Backups over 7-day test window
Expected Result	Encrypted backup is taken every 24 hours, retained for seven days, and restoration to retained points succeeds.
Actual Result	
Test Result	

5.11 Nonfunctional, Safety & Security

Test Case ID	ST_29
Module	Nonfunctional, Safety & Security
Test Case Description	Verify transaction save meets 1-second 95th percentile target over 4G (PER-1)
Preconditions	Test environment uses representative 4G connection and transaction dataset
Test Steps	1. Execute repeated saves. 2. Record save-to-confirmation latency. 3. Calculate 95th percentile.
Test Data	Representative transaction workload
Expected Result	95th-percentile transaction save completion is <=1 second.
Actual Result	
Test Result	

Test Case ID	ST_30
Module	Nonfunctional, Safety & Security
Test Case Description	Verify screens become interactive within 2s on 4G and 5s on 3G (PER-2)
Preconditions	Representative mobile/web test environment
Test Steps	1. Navigate through representative screens over 4G and 3G. 2. Measure time to interactive.
Test Data	4G and 3G network profiles
Expected Result	Screens become interactive within 2 seconds on 4G and 5 seconds on 3G.
Actual Result	
Test Result	

Test Case ID	ST_31
Module	Nonfunctional, Safety & Security
Test Case Description	Verify dashboard renders within 3 seconds of sign-in (PER-3)
Preconditions	Current-period dashboard data exists
Test Steps	1. Sign in. 2. Measure until summary figures and primary chart render.
Test Data	Representative dashboard dataset
Expected Result	Dashboard summary figures and primary chart render within 3 seconds.
Actual Result	
Test Result	

Test Case ID	ST_32
Module	Nonfunctional, Safety & Security
Test Case Description	Verify transaction list and filtering/search performance (PER-5)
Preconditions	At least 25 transactions exist
Test Steps	1. Load a 25-record page. 2. Measure filtering/search.
Test Data	25+ transactions
Expected Result	Page returns within 1 second; filtering/search returns within 2 seconds.
Actual Result	
Test Result	

Test Case ID	ST_33
Module	Nonfunctional, Safety & Security
Test Case Description	Verify 200 concurrent active users do not exceed 20% response-time degradation (PER-7)
Preconditions	Load-test environment configured
Test Steps	1. Run 200 concurrent users. 2. Compare response times with baseline.
Test Data	200 concurrent active users
Expected Result	System supports at least 200 concurrent users without degrading stated response targets by more than 20%.
Actual Result	
Test Result	

Test Case ID	ST_34
Module	Nonfunctional, Safety & Security
Test Case Description	Verify scheduler completes due rules/reminders within 10 minutes of 00:05 IST (PER-8)
Preconditions	Due recurring rules/reminders exist
Test Steps	1. Trigger scheduler for test day. 2. Measure completion.
Test Data	Due rules and reminders
Expected Result	All due recurring rules/reminders are processed within 10 minutes of 00:05 IST.
Actual Result	
Test Result	

Test Case ID	ST_35
Module	Nonfunctional, Safety & Security
Test Case Description	Verify 1000-row CSV import preview/commit performance (PER-9)
Preconditions	1000-row valid CSV is available
Test Steps	1. Import 1000 rows. 2. Measure validation/preview. 3. Confirm and measure commit.
Test Data	1000-row CSV
Expected Result	Preview/validation completes within 15 seconds and commit completes within a further 15 seconds.
Actual Result	
Test Result	

Test Case ID	ST_36
Module	Nonfunctional, Safety & Security
Test Case Description	Verify no financial advice or product recommendations are presented (SAF-1)
Preconditions	Application screens/reports are accessible
Test Steps	1. Review dashboard, reports, goals and projections for recommendation/advice content.
Test Data	Representative user data
Expected Result	System reports recorded data only and provides no investment, tax, insurance, credit advice or financial-product recommendation.
Actual Result	
Test Result	

Test Case ID	ST_37
Module	Nonfunctional, Safety & Security
Test Case Description	Verify projections and required monthly saving figures are labelled as arithmetic projections (SAF-2)
Preconditions	Dashboard/goal projection is available
Test Steps	1. Open projection and goal details. 2. Inspect explanatory text.
Test Data	Projection data
Expected Result	Projection areas state that figures are arithmetic projections from recorded data and not guarantees.
Actual Result	
Test Result	

Test Case ID	ST_38
Module	Nonfunctional, Safety & Security
Test Case Description	Verify failed transfer, goal contribution and CSV import leave no partial financial record (SAF-5)
Preconditions	Test environment can force operation failure
Test Steps	1. Force failure during each operation. 2. Compare data before/after.
Test Data	Transfer; goal contribution; CSV import with induced failure
Expected Result	Each operation either completes fully or leaves financial data unchanged; no partial record remains.
Actual Result	
Test Result	

Test Case ID	ST_39
Module	Nonfunctional, Safety & Security
Test Case Description	Verify stale balances are never displayed as current (SAF-6)
Preconditions	Test environment can interrupt recomputation
Test Steps	1. Change transaction and interrupt recomputation. 2. Observe displayed balance.
Test Data	Balance recomputation failure
Expected Result	Incorrect stale balance is not shown as current; figure is shown unavailable until recomputed.
Actual Result	
Test Result	

Test Case ID	ST_40
Module	Nonfunctional, Safety & Security
Test Case Description	Verify irreversible actions are preceded by clear warning (SAF-7)
Preconditions	Account deletion or permanent purge is available in test environment
Test Steps	1. Initiate irreversible action. 2. Inspect warning. 3. Cancel.
Test Data	Account deletion; permanent purge
Expected Result	System warns before irreversible action and identifies what cannot be undone.
Actual Result	
Test Result	

Test Case ID	ST_41
Module	Nonfunctional, Safety & Security
Test Case Description	Verify no user can access another user's financial data (SAF-8)
Preconditions	Two separate user accounts exist
Test Steps	1. Sign in as User A. 2. Attempt direct access to User B transaction/report/attachment resources. 3. Repeat with administrator.
Test Data	User A and User B data
Expected Result	No user or administrator can expose another user's financial data.
Actual Result	
Test Result	

Test Case ID	ST_42
Module	Nonfunctional, Safety & Security
Test Case Description	Verify daily backup recovery objectives (SAF-4)
Preconditions	Encrypted daily backups exist
Test Steps	1. Select retained restore point. 2. Perform restore. 3. Measure recovery time and point.
Test Data	Backup within last 7 days
Expected Result	Recovery point is within 24 hours and restoration completes within the stated 4-hour recovery time objective.
Actual Result	
Test Result	

Test Case ID	ST_43
Module	Nonfunctional, Safety & Security
Test Case Description	Verify mobile package and offline cache size limits (PER-11)
Preconditions	Mobile build is available
Test Steps	1. Inspect release package size. 2. Populate offline cache to representative maximum. 3. Measure.
Test Data	Release APK/IPA and offline data
Expected Result	Application package <=30MB and offline cache <=50MB.
Actual Result	
Test Result	

Appendix A — Requirement-to-Test Traceability

This mapping is reproduced from the PFEM Test Plan and provides a quick check that SRS requirements are represented by test cases.

Requirement ID	Test Case ID(s)
REQ-1	UT_01, UT_02
REQ-2	UT_03
REQ-3	UT_04
REQ-4	UT_05, UT_06
REQ-5	UT_76
REQ-6	UT_61
REQ-7	ST_01, UT_07
REQ-8	IT_01
REQ-9	UT_62, IT_17
REQ-10	UT_63
REQ-11	UT_64
REQ-12	UT_65
REQ-13	UT_08, UT_09, ST_02
REQ-14	UT_10, ST_02
REQ-15	UT_66
REQ-16	UT_67
REQ-17	UT_68
REQ-18	UT_12, IT_02, ST_02
REQ-19	IT_03, ST_02
REQ-20	UT_13, ST_02
REQ-21	ST_45
REQ-22	UT_14, ST_03
REQ-23	UT_15
REQ-24	IT_05, ST_03
REQ-25	UT_16, UT_69, ST_03
REQ-26	UT_17, UT_70, ST_03
REQ-27	UT_71
REQ-28	UT_19, IT_04, ST_03
REQ-29	UT_20, ST_13, ST_03
REQ-30	ST_14
REQ-31	ST_46
REQ-32	UT_72
REQ-33	ST_48
REQ-34	ST_47
REQ-35	UT_73
REQ-36	UT_21, UT_22, ST_04
REQ-37	UT_74
REQ-38	IT_21
REQ-39	IT_22
REQ-40	UT_25, ST_04
REQ-41	ST_49
REQ-42	ST_50, IT_06, ST_04
REQ-43	UT_27, UT_28, ST_05
REQ-44	UT_29, ST_05

Requirement ID	Test Case ID(s)
REQ-45	ST_15
REQ-46	IT_07, IT_08, IT_36, ST_05
REQ-47	ST_16
REQ-48	UT_32, ST_05
REQ-49	UT_75
REQ-50	ST_17
REQ-51	ST_56
REQ-52	UT_33, UT_34, ST_06
REQ-53	UT_18
REQ-54	IT_09, IT_37, ST_06
REQ-55	IT_23
REQ-56	IT_24
REQ-57	UT_77, IT_28, UT_37, IT_10, ST_06
REQ-58	ST_18
REQ-59	ST_19
REQ-60	IT_25
REQ-61	UT_39, ST_07
REQ-62	UT_41
REQ-63	IT_26
REQ-64	UT_43, ST_07
REQ-65	ST_20
REQ-66	IT_11, ST_07
REQ-67	IT_27
REQ-68	ST_21
REQ-69	UT_45, ST_08
REQ-70	UT_46, UT_80, ST_08
REQ-71	UT_78
REQ-72	ST_51
REQ-73	IT_20
REQ-74	UT_79
REQ-75	IT_29
REQ-76	ST_52
REQ-77	UT_54, UT_81, ST_09
REQ-78	ST_53
REQ-79	UT_51, UT_82, ST_10
REQ-80	UT_52, UT_53, ST_10
REQ-81	ST_54
REQ-82	IT_30
REQ-83	UT_55, ST_10
REQ-84	UT_56
REQ-85	IT_31
REQ-86	IT_14, IT_38, ST_10
REQ-87	ST_55
REQ-88	IT_15
REQ-89	UT_59, UT_60
REQ-90	IT_32

Requirement ID	Test Case ID(s)
REQ-91	IT_33
REQ-92	IT_34
REQ-93	IT_16
REQ-94	IT_35
REQ-95	ST_11
REQ-96	ST_26
REQ-97	ST_28
SEC-8	ST_12
PER-1	ST_29
PER-2	ST_30
PER-3	ST_31
PER-5	ST_32
PER-7	ST_33
PER-8	ST_34
PER-9	ST_35
PER-11	ST_43
SAF-1	ST_36
SAF-2	ST_37
SAF-4	ST_42
SAF-5	ST_38
SAF-6	ST_39
SAF-7	ST_40
SAF-8	ST_41

Appendix B — Execution Guidance

- Execute preconditions before each case or use a controlled shared test dataset.
- Record the Actual Result exactly as observed; do not replace it with the expected result.
- Mark Pass only when the Actual Result satisfies the Expected Result.
- Mark Fail when the expected behaviour is not achieved and record the corresponding defect ID.
- Retest failed cases after defect fixes and preserve the original execution evidence.
- Keep test data and execution logs consistent with the Test Plan.